PAPER_375 — UQFF Advanced Integration

Wormhole-MUGE Term | Meissner Exponential | Relativistic γ | Error Propagation

Star Magic UQFF Whitepaper Series

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Source Documents: "Compressed UQFF Equation_14May2025.docx",

"Master UQFF Resonance Equation_14May2025.docx",

"UQFF_Resonance Superconductive Universal Gravity Equation system proof set._15May2025.docx"

Abstract

> **Key UQFF calibrated constants:** κ = 5.0e-4 day^{-1}; [SSq] = 5.7e-1; H_SCm ≈ 9.9e-1; U_UA ≈ 1.0e-4; k_η = 1.0e-113; β_i ≈ 6.0e-1; G = 6.674e-11 N·m^2/kg^2

This paper presents the advanced integration of the UQFF framework combining four new mathematical formulations: (1) a wormhole-MUGE coupling term derived from the Morris-Thorne metric; (2) the exponential Meissner superconductivity model replacing the linear B/Bcrit quenching; (3) a special-relativistic Lorentz correction applied to the DPM acceleration term for high-velocity systems; and (4) a formal error propagation formalism for uncertainty quantification across all MUGE terms. These four enhancements are combined into a single Unified UQFF master equation and validated for the J1610+1811 system at v=0.99c.

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1. Wormhole-MUGE Coupling Term

The Morris-Thorne wormhole geometry (PAPER_373) introduces a new acceleration term in the MUGE resonance sum:

$$a_{\text{worm}}(r) = \frac{f_{\text{worm}} \cdot E_{\text{vac,neb}}}{b^2 + r^2}$$

where: - $f_{\text{worm}} = 10^{-10}$ (dimensionless wormhole coupling constant) - $E_{\text{vac,neb}} = 7.09 \times 10^{-36}$ J (nebular vacuum energy) - $b = 1.0$ m (wormhole throat radius) - r = evaluation radius (m)

This term encodes the gravitational contribution of a wormhole throat at distance r, modulated by the vacuum energy of the local medium.

At $r = 1$ m, $b = 1$:

$$a_{\text{worm}}(1) = \frac{10^{-10} \times 7.09 \times 10^{-36}}{1+1} = 3.545 \times 10^{-46} \text{ m/s}^2$$

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2. Meissner Exponential Superconductivity

PAPER_372 uses the linear Meissner approximation: $(1 - B/B_{\text{crit}})$.

This paper introduces the physically more accurate **exponential form** applicable to Type-II superconductors (London penetration depth model):

$$\left(1 - \frac{B}{B_{\text{crit}}}\right) \longrightarrow e^{-B/B_{\text{crit}}}$$

For the Compressed UQFF master equation, the gravitational coupling becomes:

$$g_{\text{UQFF}} = \frac{GM}{r^2} \cdot [1 + H_0 t] \cdot e^{-B/B_{\text{crit}}} \cdot [1 + F_{\text{env}}] + ...$$

The exponential form ensures monotone decay from $e^0 = 1.0$ (no field) to 0 (field well above Bcrit), without unphysical negative values that arise from the linear form when $B > B_{\text{crit}}$.

System	B/Bcrit	Linear factor	Exponential factor
SGR1745-2900	0.1	0.9	0.905
SgrA*	0.1	0.9	0.905
Student's Guide	0.1	0.9	0.905

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3. Relativistic Lorentz Correction

For high-velocity systems (e.g., J1610+1811 jet at v = 0.99c), the DPM acceleration term undergoes Lorentz suppression:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$a_{\text{DPM}} \longrightarrow \frac{a_{\text{DPM}}}{\gamma}$$

For $v = 0.99c$:

$$\gamma = \frac{1}{\sqrt{1 - 0.9801}} = \frac{1}{\sqrt{0.0199}} \approx 7.089$$

This suppresses a_{DPM} by factor ~7, reflecting that the DPM force (electromagnetic in origin) is frame-dependent in the relativistic regime. All other resonance terms retain their coordinate-frame values.

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4. Error Propagation Formalism

Uncertainties in individual MUGE terms propagate in quadrature:

$$\delta g = \sqrt{\sum_i (\delta a_i)^2}$$

where δa_i is the uncertainty in each term a_i. For a fractional error $f = 1\%$:

$$\delta a_i = f \cdot |a_i| \qquad \Rightarrow \qquad \delta g = f \cdot \sqrt{\sum_i a_i^2}$$

This provides a rigorous uncertainty bound for UQFF predictions, enabling comparison with observational error bars.

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5. Unified UQFF Master Equation (Complete Form)

Combining all prior papers (PAPER_371–375):

$$g(r,t) = \underbrace{\Bigg[\frac{GM}{r^2}(1 + H_0 t)\, e^{-B/B_{\text{crit}}}(1 + F_{\text{env}}) + \sum U_{gi} + \frac{\Lambda c^2}{3} + \frac{\hbar}{\Delta x \Delta p} \int \psi^* \hat{H} \psi \, dV \cdot \frac{2\pi}{t_H} + \rho_f V g \cdots}_{\text{Compressed UQFF (PAPER 372, Meissner exp)}}$$

$$+ \underbrace{\Bigg[\frac{a_{\text{DPM}}}{\gamma} + a_{\text{THz}} + a_{\text{vac,diff}} + a_{\text{super,freq}} + a_{\text{aether,res}} + U_{g4i} + a_{\text{quantum,freq}} + a_{\text{Aether,freq}} + a_{\text{fluid,freq}} + a_{\cdots}}_{\text{MUGE Resonance with Lorentz correction (PAPER 371)}}$$

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6. Canonical Parameter Summary

Symbol	Value	Paper
f_worm	1×10^{-10}	PAPER_375
Meissner form	exp(−B/Bcrit)	PAPER_375 (vs linear PAPER_372)
γ (v=0.99c)	≈ 7.089	PAPER_375
δg (1% error, SGR1745)	~10^{-9} × 0.01	PAPER_375

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7. Implementation

C++: `STAR_MAGIC_09SEPT_UQFF_MODULE.cpp`, namespace `UQFFAdvanced` - `compute_a_wormhole(Evac_` `b, r)` — wormhole MUGE term - `meissner_exp(B, Bcrit)` — exponential Meissner factor - `lorentz_gamma(v)` — relativistic Lorentz factor - `apply_lorentz(aDPM, v)` — Lorentz-corrected DPM - `error_propagation(delta_terms)` — quadrature error propagation - `compute_unified_UQFF(sys, res, t, v_jet, b_worm, r_worm)` — master unified function - `compute_total_uncertainty(sys, p, frac_error)` — uncertainty budget

Python: `CondensedPhysics4.py`, class `UQFFWormholeMeissnerRelativisticGammaCalculator` (CP4 #23)

WOLFRAM_TERM: `WOLFRAM_TERM_UQFF_ADVANCED`

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